Summary report of CHAEA learning styles by CHAEA³S package

This report contains the most important results of the analysis that is conducted to unveil the learning styles that are present in the group of students under study. The analysis is based on the learning styles considered by CHAEA: activist, reflector, theorist, and pragmatist. Unless otherwise stated, the uncertainties throughout the document (in parenthesis) have been obtained using a t-Student distribution with a confidence interval of 95%.

The report is structured as follows. First, Section 1 is devoted to the individual and global statistical analysis. Here, the importance of the different learning styles for each individual student can be found (both quantitatively as well as qualitatively). Then, average means and confidence intervals are presented, along with the affinities and the Probability Density Functions. Second, Section 2 discusses the principal component analysis. The eigenvalues and eigenvectors of the covariance matrix are first introduced. Subsequently, the learning styles of the students in the principal components basis set is presented. Next, a reduced dimensional representation of the data is conducted. Third, the participation ratios are presented in Section 3. The values for the original (activist, reflector, theorist, and pragmatist) and in the principal components basis sets are listed. Here, a statistical analysis of the distribution of the participation ratios is performed. Section 4 presents the clustering analysis of the data based on K-means algorithm with K=2 clusters. In this section, the relationship between the students marks, and the CHAEA learning styles and the principal components The report concludes in Section 5 with a brief summary of the main characteristics of the students, along with some recommendations.

Further information at: J. Ablanque, V. Gabaldon, P. Almendros, J. C. Losada, R. M. Benito, and F. Revuelta. CHAEA3S: A software package for comprehensive analysis of learning styles and academic performance. Computers and Education (2025).

1. Individual and global statistical analysis

In this section we present the individual and global statistical analysis of the learning styles as originally defined in CHAEA (activist, reflector, theorist, and pragmatist). This section is divided in two parts. First, the (quantitative and qualitative) importance of the different learning styles for each individual student can be found. Second, a global analysis is performed, where average means, confidence intervals, affinities, and the probability density functions can be found.

1.1 Individual analysis

Quantitative description of the learning styles for each individual student

Table 1 shows the number of points (from 0 to 20) that each student gets in CHAEA for the different learning styles.

Table 1. Points related the learning styles for each of the students.

Student	Activist	Reflector	Theorist	Pragmatist
fisicaI_2022-23_01	14.0	14.0	14.0	15.0
fisicaI_2022-23_02	13.0	18.0	12.0	10.0
fisicaI_2022-23_03	13.0	15.0	14.0	13.0
fisicaI_2022-23_04	9.0	17.0	13.0	8.0
fisicaI_2022-23_05	15.0	18.0	12.0	13.0
fisicaI_2022-23_06	17.0	13.0	16.0	12.0
fisicaI_2022-23_07	14.0	17.0	11.0	11.0
fisicaI_2022-23_08	8.0	18.0	20.0	15.0
fisicaI_2022-23_09	6.0	15.0	15.0	12.0
fisicaI_2022-23_11	14.0	16.0	12.0	14.0
fisicaI_2022-	17.0	15.0	19.0	18.0

23_12				
fisicaI_2022-23_13	17.0	15.0	13.0	17.0
fisicaI_2022-23_14	6.0	17.0	8.0	9.0
fisicaI_2022-23_15	10.0	13.0	11.0	10.0
fisicaI_2022-23_17	9.0	16.0	12.0	13.0
fisicaI_2022-23_18	9.0	19.0	14.0	13.0
fisicaI_2022-23_19	10.0	16.0	15.0	16.0
fisicaI_2022-23_20	10.0	12.0	13.0	13.0
fisicaI_2022-23_21	15.0	16.0	11.0	15.0
fisicaI_2022-23_22	15.0	13.0	11.0	19.0
fisicaI_2022-23_23	11.0	18.0	17.0	12.0
fisicaI_2022-23_24	10.0	12.0	13.0	16.0
fisicaI_2022-23_25	12.0	15.0	13.0	10.0
fisicaI_2022-23_26	13.0	16.0	11.0	11.0
fisicaI_2022-23_27	13.0	19.0	15.0	14.0
fisicaI_2022-23_30	9.0	16.0	12.0	14.0

fisicaI_2022-23_31	15.0	19.0	9.0	15.0
fisicaI_2022-23_33	10.0	9.0	11.0	14.0
fisicaI_2022-23_34	15.0	10.0	13.0	20.0
fisicaI_2022-23_35	16.0	18.0	8.0	9.0
fisicaI_2022-23_36	10.0	15.0	11.0	14.0
fisicaI_2022-23_37	8.0	15.0	12.0	13.0
fisicaI_2022-23_38	10.0	15.0	8.0	7.0
fisicaI_2022-23_39	4.0	20.0	18.0	13.0
fisicaI_2022-23_40	9.0	19.0	11.0	19.0
fisicaI_2022-23_41	11.0	15.0	16.0	16.0
fisicaI_2022-23_42	11.0	17.0	15.0	12.0
fisicaI_2022-23_43	9.0	16.0	14.0	13.0
fisicaI_2022-23_44	12.0	15.0	13.0	14.0
fisicaI_2022-23_45	14.0	13.0	15.0	12.0
fisicaI_2022-23_46	8.0	18.0	12.0	12.0
fisicaI_2022-23_47	11.0	15.0	10.0	10.0
fisicaI_2022-	8.0	19.0	19.0	15.0

23_48				
fisicaI_2022-23_49	12.0	16.0	13.0	12.0
fisicaI_2022-23_50	5.0	19.0	17.0	11.0
fisicaI_2022-23_51	2.0	16.0	14.0	7.0
fisicaI_2022-23_52	6.0	16.0	13.0	11.0
fisicaI_2022-23_53	12.0	18.0	14.0	17.0
fisicaI_2022-23_54	11.0	17.0	15.0	12.0
fisicaI_2022-23_55	12.0	16.0	15.0	10.0
fisicaI_2022-23_56	11.0	18.0	10.0	12.0
fisicaI_2022-23_57	13.0	12.0	13.0	14.0
fisicaI_2022-23_58	12.0	17.0	16.0	12.0
fisicaI_2022-23_59	5.0	15.0	13.0	8.0
fisicaI_2022-23_61	12.0	12.0	10.0	12.0
fisicaI_2022-23_62	7.0	14.0	14.0	12.0
fisicaI_2022-23_63	10.0	16.0	14.0	11.0
fisicaI_2022-23_64	7.0	13.0	8.0	8.0

fisicaI_2022-23_65	9.0	13.0	16.0	14.0
fisicaI_2022-23_66	6.0	15.0	9.0	7.0
fisicaI_2022-23_67	12.0	14.0	11.0	15.0
fisicaI_2022-23_68	13.0	15.0	11.0	9.0
fisicaI_2022-23_69	15.0	14.0	12.0	12.0
fisicaI_2022-23_70	13.0	13.0	6.0	7.0
fisicaI_2022-23_71	0.0	0.0	0.0	0.0
fisicaI_2022-23_72	13.0	17.0	16.0	12.0
fisicaI_2022-23_73	12.0	17.0	14.0	14.0
fisicaI_2022-23_74	6.0	15.0	15.0	7.0
fisicaI_2022-23_75	12.0	12.0	14.0	10.0
fisicaI_2022-23_76	14.0	10.0	7.0	10.0
fisicaI_2023-24_01	13.0	16.0	10.0	13.0
fisicaI_2023-24_02	10.0	17.0	18.0	19.0
fisicaI_2023-24_03	14.0	17.0	18.0	15.0
fisicaI_2023-24_04	17.0	16.0	12.0	13.0
fisicaI_2023-	17.0	11.0	9.0	9.0

24_05				
fisicaI_2023-24_06	9.0	20.0	17.0	14.0
fisicaI_2023-24_07	8.0	18.0	17.0	14.0
fisicaI_2023-24_08	15.0	11.0	9.0	14.0
fisicaI_2023-24_09	13.0	15.0	14.0	16.0
fisicaI_2023-24_10	9.0	17.0	13.0	13.0
fisicaI_2023-24_11	7.0	16.0	15.0	11.0
fisicaI_2023-24_12	9.0	13.0	16.0	14.0
fisicaI_2023-24_13	10.0	18.0	12.0	8.0
fisicaI_2023-24_14	11.0	15.0	13.0	12.0
fisicaI_2023-24_15	11.0	20.0	15.0	10.0
fisicaI_2023-24_16	8.0	19.0	15.0	13.0
fisicaI_2023-24_17	12.0	17.0	15.0	10.0
fisicaI_2023-24_18	9.0	16.0	16.0	18.0
fisicaI_2023-24_20	3.0	17.0	18.0	12.0
fisicaI_2023-24_21	7.0	17.0	15.0	11.0

fisicaI_2023-24_22	10.0	9.0	15.0	9.0
fisicaI_2023-24_23	9.0	15.0	14.0	11.0
fisicaI_2023-24_24	14.0	13.0	8.0	15.0
fisicaI_2023-24_25	16.0	11.0	15.0	15.0
fisicaI_2023-24_26	8.0	18.0	17.0	10.0
fisicaI_2023-24_27	15.0	18.0	18.0	18.0
fisicaI_2023-24_28	6.0	20.0	17.0	13.0
fisicaI_2023-24_29	6.0	1.0	3.0	6.0
fisicaI_2023-24_30	14.0	17.0	15.0	13.0
fisicaI_2023-24_31	17.0	17.0	15.0	20.0
fisicaI_2023-24_32	12.0	11.0	18.0	11.0
fisicaI_2023-24_33	15.0	17.0	13.0	13.0
fisicaI_2023-24_34	16.0	13.0	14.0	18.0
fisicaI_2023-24_35	11.0	19.0	12.0	12.0
fisicaI_2023-24_36	12.0	16.0	15.0	16.0
fisicaI_2023-24_37	14.0	13.0	7.0	11.0
fisicaI_2023-	6.0	17.0	16.0	15.0

24_38				
fisicaI_2023-24_39	15.0	19.0	14.0	14.0
fisicaI_2023-24_40	10.0	18.0	14.0	13.0
fisicaI_2023-24_41	2.0	17.0	13.0	12.0
fisicaI_2023-24_42	15.0	12.0	10.0	13.0
fisicaI_2023-24_43	9.0	19.0	15.0	11.0
fisicaI_2023-24_44	3.0	17.0	15.0	9.0
fisicaI_2023-24_45	8.0	18.0	16.0	17.0
fisicaI_2023-24_46	5.0	20.0	14.0	11.0
fisicaI_2023-24_47	10.0	13.0	15.0	14.0
fisicaI_2023-24_48	16.0	13.0	11.0	16.0
fisicaI_2023-24_49	9.0	15.0	12.0	14.0
fisicaI_2023-24_50	5.0	17.0	16.0	12.0
fisicaI_2023-24_51	13.0	17.0	10.0	12.0
fisicaI_2023-24_52	11.0	15.0	15.0	14.0
fisicaI_2023-24_53	11.0	18.0	16.0	14.0

fisicaI_2023-24_54	11.0	15.0	14.0	13.0
fisicaI_2023-24_55	13.0	12.0	16.0	17.0
fisicaI_2023-24_56	12.0	13.0	15.0	10.0
fisicaI_2023-24_57	11.0	15.0	16.0	13.0
fisicaI_2023-24_58	9.0	19.0	18.0	16.0
fisicaI_2023-24_59	14.0	18.0	16.0	16.0
fisicaI_2023-24_60	14.0	18.0	18.0	18.0

Qualitative description of the learning styles for each individual student

Table 2 shows the individual qualitative tendency (very low, low, moderate, high, or very high) that each student has towards each of the learning styles.

Table 2. Tendency towards the learning styles for each of the students.

Student	Activist	Reflector	Theorist	Pragmatist
fisicaI_2022-23_01	High	Moderate	High	High
fisicaI_2022-23_02	High	High	Moderate	Low
fisicaI_2022-23_03	High	Moderate	High	Moderate
fisicaI_2022-23_04	Moderate	Moderate	Moderate	Very low
fisicaI_2022-23_05	Very high	High	Moderate	Moderate
fisicaI_2022-23_06	Very high	Low	Very high	Moderate
fisicaI_2022-23_07	High	Moderate	Moderate	Moderate
fisicaI_2022-23_08	Low	High	Very high	High
fisicaI_2022-23_09	Very low	Moderate	High	Moderate
fisicaI_2022-23_11	High	Moderate	Moderate	High
fisicaI_2022-23_12	Very high	Moderate	Very high	Very high
fisicaI_2022-23_13	Very high	Moderate	Moderate	Very high
fisicaI_2022-23_14	Very low	Moderate	Low	Low

fisicaI_2022-23_15	Moderate	Low	Moderate	Low
fisicaI_2022-23_17	Moderate	Moderate	Moderate	Moderate
fisicaI_2022-23_18	Moderate	High	High	Moderate
fisicaI_2022-23_19	Moderate	Moderate	High	Very high
fisicaI_2022-23_20	Moderate	Low	Moderate	Moderate
fisicaI_2022-23_21	Very high	Moderate	Moderate	High
fisicaI_2022-23_22	Very high	Low	Moderate	Very high
fisicaI_2022-23_23	Moderate	High	Very high	Moderate
fisicaI_2022-23_24	Moderate	Low	Moderate	Very high
fisicaI_2022-23_25	Moderate	Moderate	Moderate	Low
fisicaI_2022-23_26	High	Moderate	Moderate	Moderate
fisicaI_2022-23_27	High	High	High	High
fisicaI_2022-23_30	Moderate	Moderate	Moderate	High
fisicaI_2022-23_31	Very high	High	Low	High
fisicaI_2022-23_33	Moderate	Very low	Moderate	High
fisicaI_2022-23_34	Very high	Very low	Moderate	Very high
fisicaI_2022-	Very high	High	Low	Low

23_35				
fisicaI_2022-23_36	Moderate	Moderate	Moderate	High
fisicaI_2022-23_37	Low	Moderate	Moderate	Moderate
fisicaI_2022-23_38	Moderate	Moderate	Low	Very low
fisicaI_2022-23_39	Very low	Very high	Very high	Moderate
fisicaI_2022-23_40	Moderate	High	Moderate	Very high
fisicaI_2022-23_41	Moderate	Moderate	Very high	Very high
fisicaI_2022-23_42	Moderate	Moderate	High	Moderate
fisicaI_2022-23_43	Moderate	Moderate	High	Moderate
fisicaI_2022-23_44	Moderate	Moderate	Moderate	High
fisicaI_2022-23_45	High	Low	High	Moderate
fisicaI_2022-23_46	Low	High	Moderate	Moderate
fisicaI_2022-23_47	Moderate	Moderate	Moderate	Low
fisicaI_2022-23_48	Low	High	Very high	High
fisicaI_2022-23_49	Moderate	Moderate	Moderate	Moderate
fisicaI_2022-23_50	Very low	High	Very high	Moderate

fisicaI_2022-23_51	Very low	Moderate	High	Very low
fisicaI_2022-23_52	Very low	Moderate	Moderate	Moderate
fisicaI_2022-23_53	Moderate	High	High	Very high
fisicaI_2022-23_54	Moderate	Moderate	High	Moderate
fisicaI_2022-23_55	Moderate	Moderate	High	Low
fisicaI_2022-23_56	Moderate	High	Moderate	Moderate
fisicaI_2022-23_57	High	Low	Moderate	High
fisicaI_2022-23_58	Moderate	Moderate	Very high	Moderate
fisicaI_2022-23_59	Very low	Moderate	Moderate	Very low
fisicaI_2022-23_61	Moderate	Low	Moderate	Moderate
fisicaI_2022-23_62	Low	Moderate	High	Moderate
fisicaI_2022-23_63	Moderate	Moderate	High	Moderate
fisicaI_2022-23_64	Low	Low	Low	Very low
fisicaI_2022-23_65	Moderate	Low	Very high	High
fisicaI_2022-23_66	Very low	Moderate	Low	Very low
fisicaI_2022-23_67	Moderate	Moderate	Moderate	High
fisicaI_2022-	High	Moderate	Moderate	Low

23_68				
fisicaI_2022-23_69	Very high	Moderate	Moderate	Moderate
fisicaI_2022-23_70	High	Low	Very low	Very low
fisicaI_2022-23_71	Very low	Very low	Very low	Very low
fisicaI_2022-23_72	High	Moderate	Very high	Moderate
fisicaI_2022-23_73	Moderate	Moderate	High	High
fisicaI_2022-23_74	Very low	Moderate	High	Very low
fisicaI_2022-23_75	Moderate	Low	High	Low
fisicaI_2022-23_76	High	Very low	Low	Low
fisicaI_2023-24_01	High	Moderate	Moderate	Moderate
fisicaI_2023-24_02	Moderate	Moderate	Very high	Very high
fisicaI_2023-24_03	High	Moderate	Very high	High
fisicaI_2023-24_04	Very high	Moderate	Moderate	Moderate
fisicaI_2023-24_05	Very high	Low	Low	Low
fisicaI_2023-24_06	Moderate	Very high	Very high	High
fisicaI_2023-24_07	Low	High	Very high	High

fisicaI_2023-24_08	Very high	Low	Low	High
fisicaI_2023-24_09	High	Moderate	High	Very high
fisicaI_2023-24_10	Moderate	Moderate	Moderate	Moderate
fisicaI_2023-24_11	Low	Moderate	High	Moderate
fisicaI_2023-24_12	Moderate	Low	Very high	High
fisicaI_2023-24_13	Moderate	High	Moderate	Very low
fisicaI_2023-24_14	Moderate	Moderate	Moderate	Moderate
fisicaI_2023-24_15	Moderate	Very high	High	Low
fisicaI_2023-24_16	Low	High	High	Moderate
fisicaI_2023-24_17	Moderate	Moderate	High	Low
fisicaI_2023-24_18	Moderate	Moderate	Very high	Very high
fisicaI_2023-24_20	Very low	Moderate	Very high	Moderate
fisicaI_2023-24_21	Low	Moderate	High	Moderate
fisicaI_2023-24_22	Moderate	Very low	High	Low
fisicaI_2023-24_23	Moderate	Moderate	High	Moderate
fisicaI_2023-24_24	High	Low	Low	High
fisicaI_2023-	Very high	Low	High	High

24_25				
fisicaI_2023-24_26	Low	High	Very high	Low
fisicaI_2023-24_27	Very high	High	Very high	Very high
fisicaI_2023-24_28	Very low	Very high	Very high	Moderate
fisicaI_2023-24_29	Very low	Very low	Very low	Very low
fisicaI_2023-24_30	High	Moderate	High	Moderate
fisicaI_2023-24_31	Very high	Moderate	High	Very high
fisicaI_2023-24_32	Moderate	Low	Very high	Moderate
fisicaI_2023-24_33	Very high	Moderate	Moderate	Moderate
fisicaI_2023-24_34	Very high	Low	High	Very high
fisicaI_2023-24_35	Moderate	High	Moderate	Moderate
fisicaI_2023-24_36	Moderate	Moderate	High	Very high
fisicaI_2023-24_37	High	Low	Low	Moderate
fisicaI_2023-24_38	Very low	Moderate	Very high	High
fisicaI_2023-24_39	Very high	High	High	High
fisicaI_2023-24_40	Moderate	High	High	Moderate

fisicaI_2023-24_41	Very low	Moderate	Moderate	Moderate
fisicaI_2023-24_42	Very high	Low	Moderate	Moderate
fisicaI_2023-24_43	Moderate	High	High	Moderate
fisicaI_2023-24_44	Very low	Moderate	High	Low
fisicaI_2023-24_45	Low	High	Very high	Very high
fisicaI_2023-24_46	Very low	Very high	High	Moderate
fisicaI_2023-24_47	Moderate	Low	High	High
fisicaI_2023-24_48	Very high	Low	Moderate	Very high
fisicaI_2023-24_49	Moderate	Moderate	Moderate	High
fisicaI_2023-24_50	Very low	Moderate	Very high	Moderate
fisicaI_2023-24_51	High	Moderate	Moderate	Moderate
fisicaI_2023-24_52	Moderate	Moderate	High	High
fisicaI_2023-24_53	Moderate	High	Very high	High
fisicaI_2023-24_54	Moderate	Moderate	High	Moderate
fisicaI_2023-24_55	High	Low	Very high	Very high
fisicaI_2023-24_56	Moderate	Low	High	Low
fisicaI_2023-	Moderate	Moderate	Very high	Moderate

24_57				
fisicaI_2023-24_58	Moderate	High	Very high	Very high
fisicaI_2023-24_59	High	High	Very high	Very high
fisicaI_2023-24_60	High	High	Very high	Very high

1.2 Global analysis

Global tendencies towards the learning styles of the students (I)

Table 3 presents the number and percentage of students with the same tendency (very low, low, moderate, high, or very high) towards each of the learning styles. The bottom line shows the average tendency. These results are also shown as a barr graphic in Fig. 1.

Table 3. Number and percentage of students with the same tendency towards each of the learning styles.

Tendency	Very low		Low		Moderate		High		Very high	
	No.	%	No.	%	No.	%	No.	%	No.	%
Activist	18	14.0	12	9.3	55	42.6	23	17.8	21	16.3
Reflector	6	4.7	25	19.4	65	50.4	28	21.7	5	3.9
Theorist	3	2.3	11	8.5	44	34.1	40	31.0	31	24.0
Pragmatist	11	8.5	17	13.2	51	39.5	28	21.7	22	17.1
Total	38	10.0	65	14.1	215	42.5	119	24.1	79	18.8

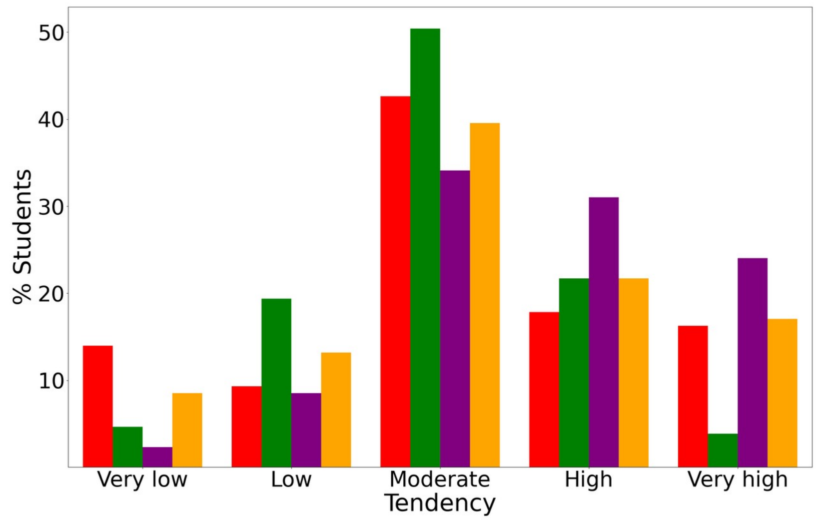

Figure 1. Bar graphic showing the percentage of students with a very low (leftmost), low (left), moderate (middle), high (right) and very high (rightmost) tendency towards the activist (red), reflector (green), theorist (purple) and pragmatist (orange) learning styles. The shown results correspond to the values of Table 3.

Global tendencies towards the learning styles of the students (II)

Table 4 shows the average mean and the uncertainties for the different learning styles, along with the corresponding qualitative tendency, and the corresponding affinity. For the shake of clarity, the average profile of the students given by the average means and the corresponding uncertainties is shown in Fig. 2, while the affinities are represented as a barr graphic in Fig. 3.

Table 4. Numerical average values of each learning style with the uncertainties (in parenthesis) along with their qualitative tendency, and the corresponding affinity.

Learning style	Average mean (Uncertainty)	Tendency	Affinity (%)
Activist	10.8(0.6)	Moderate	76.7
Reflector	15.5(0.5)	Moderate	76.0
Theorist	13.4(0.5)	Moderate/ High	89.1
Pragmatist	12.7(0.5)	Moderate	78.3

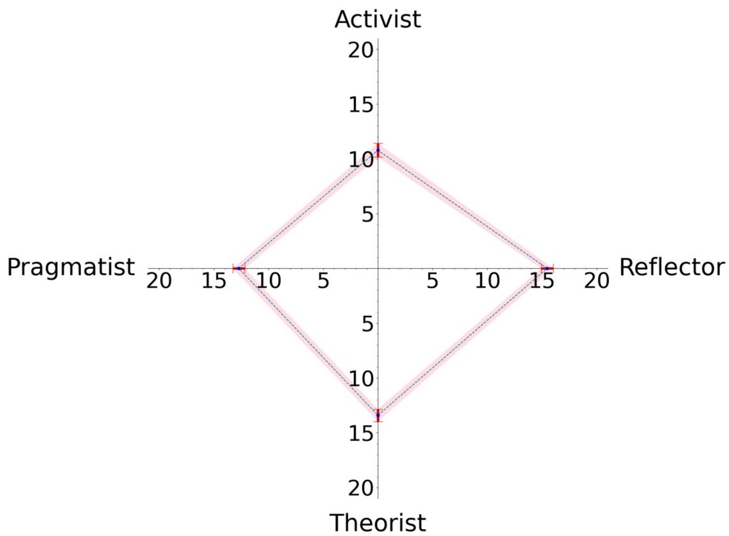

Figure 2. Average profile of the learning styles of the students.

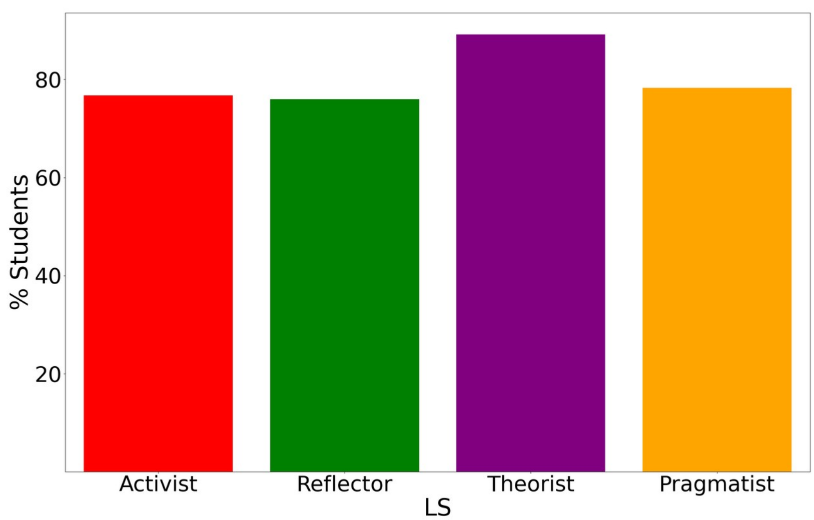

Figure 3. Affinity of the learning styles. The bar graphic gives the percentage of students that have a noticeable tendency towards the activist (red), reflector (green), theorist (purple) and pragmatist (orange) learning styles.

Figure 4 shows histograms with the probability distributions of the results of Table 1. The continuous lines show fittings with Weibull distributions with the parameters shown in Table 5. These fittings have been performed on the corresponding cumulative distributions given by the staircases shown in Figure 5.

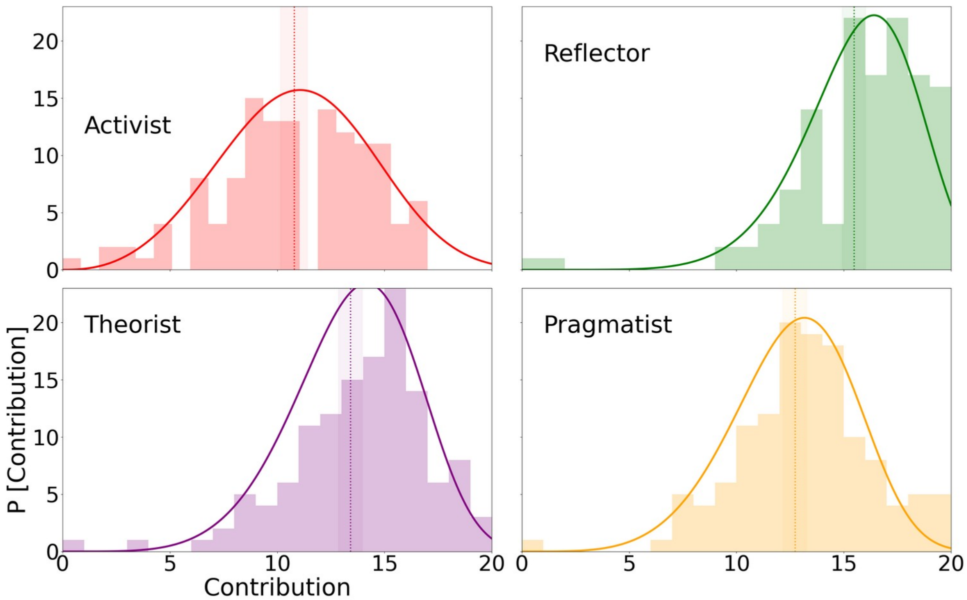

Figure 4. Histograms showing the probability distributions for the activist (red), reflector (green), theorist (purple), and pragmatist (yellow) learning styles. The continuous lines show fittings provided by the Weibull distributions with the parameters contained in Table 5. The vertical dotted lines mark the average values, and the shaded areas around them the corresponding confidence intervals.

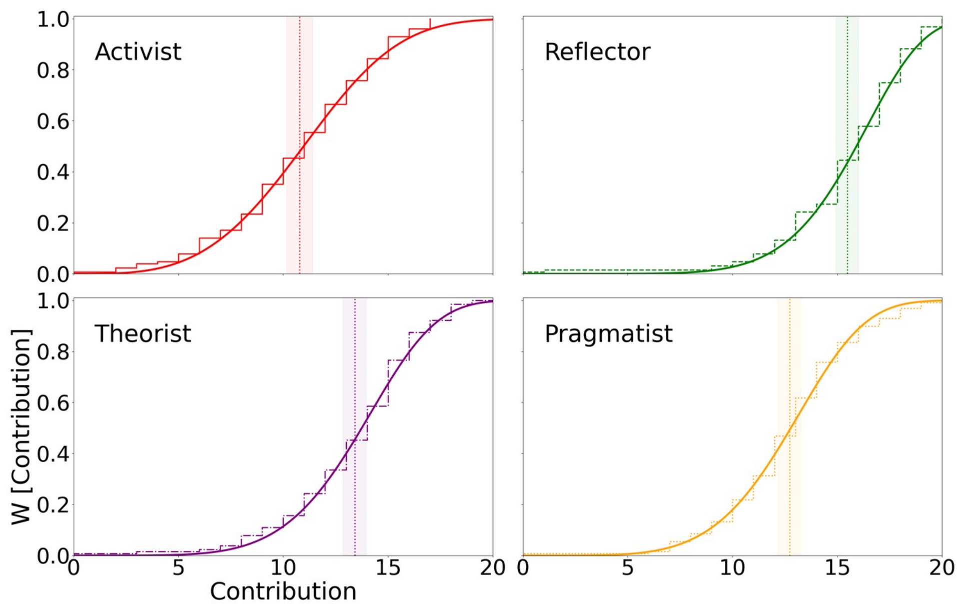

Figure 5. Same as Fig. 4 for the cumulative distributions (staircases).

Table 5. Parameters of the Weibull distributions that fit the probability distributions (histograms) shown in Figs. 4 and 5. The location parameter is set equal to $\theta = 0$.

Learning style	α	k
Activist	12.2	3.48
Reflector	16.78	6.97
Theorist	14.7	5.48
Pragmatist	13.74	5.12

1.3 Summary of the learning styles of the professors

In this section we present the learning styles of the professors as originally defined in CHAEA (activist, reflector, theorist, and pragmatist), and compare them to those of the students.

Quantitative description of the learning styles for each individual professor

Table 6 shows the number of points (from 0 to 20) that each professor gets in CHAEA for the different learning styles.

Table 6. Points related the learning styles for each of the professors.

Educator	Activist	Reflector	Theorist	Pragmatist
Chemistry	12.0	12.0	15.0	13.0
Physics	16.0	13.0	11.0	14.0

Qualitative description of the learning styles for each individual professor

Table 7 shows the individual qualitative tendency (very low, low, moderate, high, or very high) that each instructor has towards each of the learning styles.

Table 7. Tendency towards the learning styles for each of the professors.

Educator	Activist	Reflector	Theorist	Pragmatist
Chemistry	Moderate	Low	High	Moderate
Physics	Very high	Low	Moderate	High

Comparison of the learning-styles profile of the students and of each individual professor

Figures 6 to 7 show the learning style profiles of the professors superimposed with the average profile of the students shown in Fig. 2.

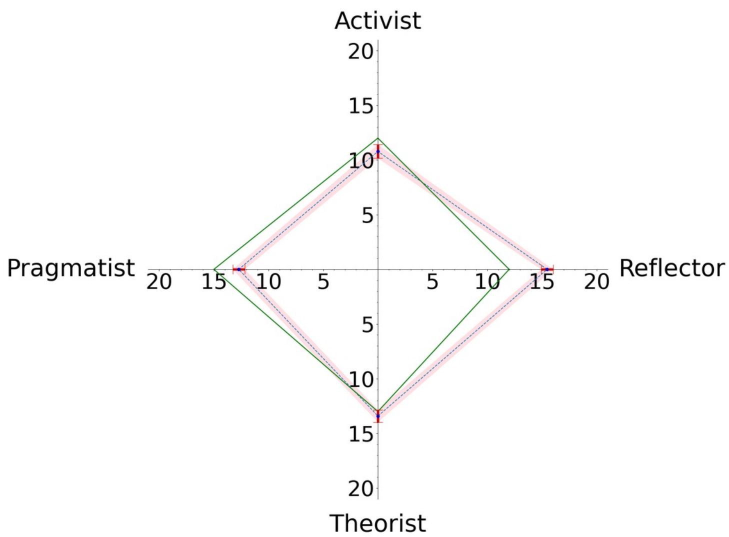

Figure 6. Profile of the learning styles of the professor "Chemistry" (green continuous line) superimposed with the average profile of the students shown in Fig. 2.

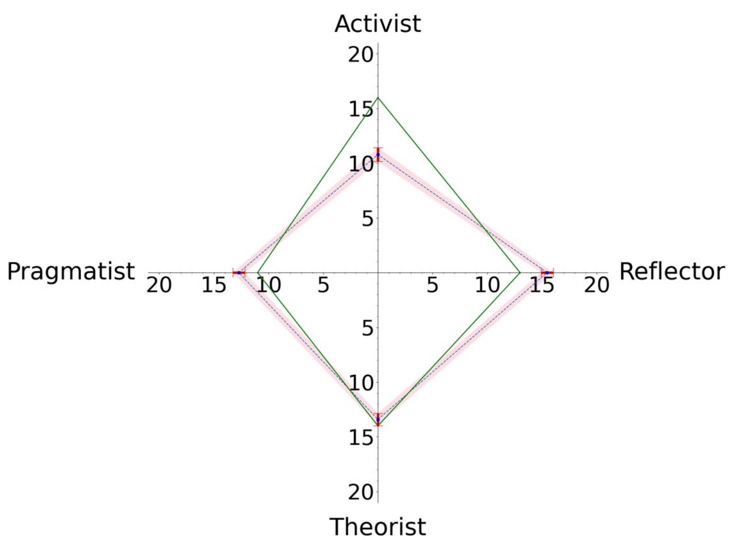

Figure 7. Same as Fig. 6 for the professor of Physics.

2. Principal component analysis

This section presents the principal component analysis of the learning styles. It is organized as follows. First, the eigenvalues and eigenvectors of the covariance matrix are presented. Second, the description of the learning styles of the students in the principal components basis set is discussed. Finally, a reduced dimensional representation of the data is conducted.

2.1 Structure of the principal components

Eigenvalues and dispersion of the covariance matrix

In this section, we discuss the essentials of the principal components. For this purpose, Table 8 shows the four eigenvalues of the covariance matrix. Here, not only their values are listed but also their contribution to the total dispersion (given by the trace of the covariance matrix $tr(K)=43.09$, which equals the sum of all the eigenvalues) as a percentage. Likewise, the dispersion accounted by solely the principal component with the largest eigenvalue $\Sigma_0(\%)=\lambda_0*100/tr(K)$, and by combining the principal components with the two, three, and four largest eigenvalues $\Sigma_1(\%)=(\lambda_0+\lambda_1)*100/tr(K)$, $\Sigma_2(\%)=(\lambda_0+\lambda_1+\lambda_2)*100/tr(K)$, and by four $\Sigma_3(\%)=(\lambda_0+\lambda_1+\lambda_2+\lambda_3)*100/tr(K)=100\%$, respectively, are given.

Table 8. Eigenvalues of the covariance matrix given by their corresponding values λ_i and as a percentage of the total dispersion, and percentage of total dispersion Σ_i accounted by combination of the principal components with the eigenvalues λ_j, being $j\leq i$.

Principal component (i)	λ_i	$\lambda_i(\%)$	$\Sigma_i(\%)$
0	19.08	44.3	44.3
1	14.93	34.6	78.9
2	5.64	13.1	92.0
3	3.44	8.0	100.0

Eigenvectors of the covariance matrix

Table 9 shows the structure of the eigenvectors of the covariance matrix in the basis set of CHAEA learning styles. Each cell contains the percentage of the eigenvector of the principal component 0, 1, 2, and 3 that is projected on the corresponding learning style (activist, reflector, theorist, and pragmatist).

Table 9. Structure (as percentages) of the eigenvectors of the covariance matrix in the basis set of CHAEA learning styles.

Principal component	Activist	Reflector	Theorist	Pragmatist
0	11.7	21.0	31.6	35.6
1	64.8	16.3	14.6	4.3
2	11.3	56.7	11.2	20.8
3	12.1	6.0	42.6	39.2

2.2 Individual analysis of the principal components

Quantitative description of the students in the basis set of the principal components

Table 10 shows (as percentages) the structure of the learning styles in the basis set formed by the principal components. The table also includes the percentage of the learning styles of each student that is described by combining the two (sum of the percentages for the principal components 0 and 1) or three (sum of the percentages for the principal components 0, 1, and 2) principal components with the largest eigenvalues. Recall that when the four principal components are considered, 100% of the learning style of the student is reproduced.

Table 10. Structure of the learning styles of each of the students in the basis set of principal components.

Student	0	1	2	3	0+1	0+1+2
fisicaI_2022-23_01	24.84	65.25	8.84	1.07	90.09	98.93
fisicaI_2022-23_02	1.21	2.64	91.92	4.23	3.85	95.77
fisicaI_2022-23_03	19.44	58.8	0.08	21.68	78.24	78.32
fisicaI_2022-23_04	31.32	29.57	28.86	10.25	60.89	89.75
fisicaI_2022-23_05	14.87	33.68	51.21	0.23	48.56	99.77
fisicaI_2022-23_06	7.85	45.51	0.18	46.46	53.36	53.54
fisicaI_2022-23_07	1.58	30.01	68.12	0.29	31.59	99.71
fisicaI_2022-23_08	44.29	44.82	8.26	2.63	89.12	97.37
fisicaI_2022-23_09	7.4	74.62	17.96	0.01	82.02	99.99
fisicaI_2022-23_11	12.1	71.05	13.11	3.74	83.15	96.26
fisicaI_2022-23_12	68.72	17.65	6.6	7.03	86.37	92.97
fisicaI_202	31.27	67.94	0.01	0.77	99.22	99.23

2-23_13						
fisicaI_2022-23_14	56.28	14.74	13.59	15.39	71.02	84.61
fisicaI_2022-23_15	96.27	2.62	0.03	1.08	98.89	98.92
fisicaI_2022-23_17	17.93	20.0	0.36	61.7	37.94	38.3
fisicaI_2022-23_18	14.11	57.04	18.69	10.16	71.15	89.84
fisicaI_2022-23_19	56.32	4.17	25.31	14.2	60.49	85.8
fisicaI_2022-23_20	28.67	7.51	63.67	0.14	36.18	99.86
fisicaI_2022-23_21	9.83	72.07	8.61	9.49	81.9	90.51
fisicaI_2022-23_22	10.57	63.36	9.05	17.02	73.92	82.98
fisicaI_2022-23_23	40.11	28.2	6.15	25.53	68.32	74.47
fisicaI_2022-23_24	0.08	10.97	76.08	12.88	11.04	87.12
fisicaI_2022-23_25	29.55	6.26	22.1	42.09	35.81	57.91
fisicaI_2022-23_26	13.68	32.65	53.51	0.16	46.34	99.84
fisicaI_2022-23_27	75.6	0.0	24.3	0.1	75.6	99.9
fisicaI_2022-23_30	2.24	10.04	1.4	86.32	12.28	13.68
fisicaI_2022-23_31	6.87	31.1	37.06	24.98	37.97	75.02

fisicaI_2022-23_33	29.33	20.08	48.35	2.25	49.4	97.75
fisicaI_2022-23_34	9.21	52.42	34.34	4.03	61.64	95.97
fisicaI_2022-23_35	7.01	26.1	66.89	0.0	33.11	100.0
fisicaI_2022-23_36	14.0	6.73	1.9	77.37	20.73	22.63
fisicaI_2022-23_37	32.15	21.04	8.95	37.87	53.18	62.13
fisicaI_2022-23_38	76.7	0.31	22.97	0.02	77.01	99.98
fisicaI_2022-23_39	7.14	92.05	0.33	0.49	99.19	99.51
fisicaI_2022-23_40	18.99	0.66	0.0	80.35	19.65	19.65
fisicaI_2022-23_41	60.36	0.02	39.46	0.17	60.38	99.83
fisicaI_2022-23_42	28.24	26.46	19.2	26.11	54.7	73.89
fisicaI_2022-23_43	0.43	84.88	7.16	7.53	85.31	92.47
fisicaI_2022-23_44	15.28	72.34	4.4	7.98	87.62	92.02
fisicaI_2022-23_45	0.95	41.07	4.95	53.03	42.02	46.97
fisicaI_2022-23_46	6.22	49.53	18.77	25.48	55.75	74.52
fisicaI_2022-23_47	70.32	6.3	22.81	0.57	76.62	99.43
fisicaI_2022-23_48	47.17	50.07	2.5	0.26	97.24	99.74
fisicaI_202	0.0	24.5	66.16	9.34	24.5	90.66

2-23_49						
fisicaI_2022-23_50	0.64	98.75	0.14	0.47	99.38	99.53
fisicaI_2022-23_51	30.96	68.47	0.02	0.55	99.43	99.45
fisicaI_2022-23_52	26.8	69.18	0.32	3.7	95.98	96.3
fisicaI_2022-23_53	75.03	1.47	0.1	23.4	76.5	76.6
fisicaI_2022-23_54	28.24	26.46	19.2	26.11	54.7	73.89
fisicaI_2022-23_55	0.04	1.4	19.67	78.88	1.45	21.12
fisicaI_2022-23_56	6.69	0.51	63.86	28.94	7.21	71.06
fisicaI_2022-23_57	0.48	69.33	28.55	1.64	69.81	98.36
fisicaI_2022-23_58	41.58	5.44	9.47	43.51	47.02	56.49
fisicaI_2022-23_59	49.04	49.85	0.0	1.11	98.89	98.89
fisicaI_2022-23_61	48.33	48.59	2.11	0.97	96.92	99.03
fisicaI_2022-23_62	24.69	46.06	29.17	0.08	70.75	99.92
fisicaI_2022-23_63	12.43	48.4	12.4	26.77	60.83	73.23
fisicaI_2022-23_64	95.46	1.29	0.95	2.3	96.75	97.7
fisicaI_2022-23_65	1.28	7.65	86.74	4.33	8.93	95.67

fisicaI_2022-23_66	79.78	13.3	5.97	0.95	93.08	99.05
fisicaI_2022-23_67	0.44	60.12	5.94	33.5	60.56	66.5
fisicaI_2022-23_68	37.15	18.18	33.76	10.92	55.33	89.08
fisicaI_2022-23_69	0.92	85.34	5.48	8.25	86.27	91.75
fisicaI_2022-23_70	64.17	19.83	15.98	0.02	84.0	99.98
fisicaI_2022-23_71	96.35	0.0	3.57	0.08	96.35	99.92
fisicaI_2022-23_72	42.66	0.01	12.76	44.58	42.67	55.42
fisicaI_2022-23_73	84.39	2.83	10.46	2.32	87.21	97.68
fisicaI_2022-23_74	32.82	51.01	0.02	16.15	83.83	83.85
fisicaI_2022-23_75	28.62	11.88	6.33	53.17	40.5	46.83
fisicaI_2022-23_76	49.61	50.26	0.13	0.0	99.86	100.0
fisicaI_2023-24_01	3.33	50.89	27.49	18.28	54.23	81.72
fisicaI_2023-24_02	72.07	4.53	19.45	3.96	76.59	96.04
fisicaI_2023-24_03	84.7	1.24	0.3	13.76	85.94	86.24
fisicaI_2023-24_04	7.41	70.86	19.57	2.16	78.27	97.84
fisicaI_2023-24_05	23.25	64.83	3.91	8.01	88.08	91.99
fisicaI_202	47.23	49.92	2.75	0.09	97.15	99.91

3-24_06						
fisicaI_2023-24_07	31.16	66.5	2.33	0.01	97.66	99.99
fisicaI_2023-24_08	9.17	86.82	1.9	2.11	95.99	97.89
fisicaI_2023-24_09	49.36	36.43	10.46	3.75	85.79	96.25
fisicaI_2023-24_10	0.01	58.58	5.45	35.96	58.59	64.04
fisicaI_2023-24_11	7.02	88.83	1.93	2.22	95.85	97.78
fisicaI_2023-24_12	1.28	7.65	86.74	4.33	8.93	95.67
fisicaI_2023-24_13	23.67	13.99	58.16	4.18	37.67	95.82
fisicaI_2023-24_14	67.37	14.47	3.5	14.66	81.85	85.34
fisicaI_2023-24_15	6.61	26.16	57.66	9.56	32.77	90.44
fisicaI_2023-24_16	13.05	77.86	4.94	4.15	90.91	95.85
fisicaI_2023-24_17	1.08	4.76	37.47	56.69	5.84	43.31
fisicaI_2023-24_18	47.25	6.24	31.78	14.73	53.48	85.27
fisicaI_2023-24_20	0.04	91.35	8.46	0.15	91.39	99.85
fisicaI_2023-24_21	2.4	96.72	0.08	0.81	99.12	99.19
fisicaI_2023-24_22	35.37	0.61	26.78	37.24	35.98	62.76

fisicaI_2023-24_23	34.57	49.76	2.01	13.65	84.34	86.35
fisicaI_2023-24_24	5.77	73.72	0.0	20.51	79.49	79.49
fisicaI_2023-24_25	7.2	62.74	18.46	11.6	69.94	88.4
fisicaI_2023-24_26	1.05	78.45	2.95	17.55	79.5	82.45
fisicaI_2023-24_27	95.03	4.05	0.53	0.4	99.07	99.6
fisicaI_2023-24_28	12.21	86.73	0.4	0.66	98.95	99.34
fisicaI_2023-24_29	85.27	5.4	9.2	0.13	90.67	99.87
fisicaI_2023-24_30	53.58	13.29	16.19	16.94	66.87	83.06
fisicaI_2023-24_31	67.45	29.06	0.39	3.1	96.51	96.9
fisicaI_2023-24_32	0.01	1.0	30.21	68.78	1.01	31.22
fisicaI_2023-24_33	21.23	44.03	32.65	2.09	65.27	97.91
fisicaI_2023-24_34	27.78	59.86	11.96	0.4	87.64	99.6
fisicaI_2023-24_35	1.46	4.92	83.06	10.56	6.38	89.44
fisicaI_2023-24_36	81.91	4.71	9.91	3.46	86.62	96.54
fisicaI_2023-24_37	35.95	53.28	7.63	3.14	89.23	96.86
fisicaI_2023-24_38	9.52	66.89	15.06	8.52	76.41	91.48
fisicaI_202	53.62	12.59	33.68	0.11	66.21	99.89

3-24_39						
fisicaI_2023-24_40	25.97	44.45	23.3	6.28	70.42	93.72
fisicaI_2023-24_41	10.99	73.59	2.23	13.19	84.58	86.81
fisicaI_2023-24_42	8.69	91.22	0.08	0.02	99.9	99.98
fisicaI_2023-24_43	3.61	69.24	25.28	1.87	72.85	98.13
fisicaI_2023-24_44	13.67	86.12	0.11	0.1	99.79	99.9
fisicaI_2023-24_45	45.42	28.83	8.75	17.0	74.25	83.0
fisicaI_2023-24_46	0.63	87.18	7.36	4.83	87.81	95.17
fisicaI_2023-24_47	0.6	0.01	96.41	2.98	0.61	97.02
fisicaI_2023-24_48	3.15	92.67	1.27	2.9	95.83	97.1
fisicaI_2023-24_49	10.45	2.76	16.21	70.59	13.2	29.41
fisicaI_2023-24_50	0.15	95.58	4.13	0.14	95.73	99.86
fisicaI_2023-24_51	4.08	27.79	58.36	9.77	31.87	90.23
fisicaI_2023-24_52	51.77	0.01	44.09	4.13	51.78	95.87
fisicaI_2023-24_53	80.68	16.63	1.88	0.81	97.32	99.19
fisicaI_2023-24_54	18.15	5.66	53.02	23.16	23.82	76.84

fisicaI_2023-24_55	24.04	22.68	52.32	0.96	46.72	99.04
fisicaI_2023-24_56	11.96	3.7	3.15	81.19	15.66	18.81
fisicaI_2023-24_57	31.06	4.53	23.09	41.32	35.59	58.68
fisicaI_2023-24_58	64.82	32.55	2.0	0.62	97.37	99.38
fisicaI_2023-24_59	94.14	4.67	1.14	0.06	98.81	99.94
fisicaI_2023-24_60	97.27	1.28	1.4	0.06	98.55	99.94

2.3 Global analysis of the principal components

Figure 8 shows a four-dimensional representation of the learning styles of the students as a function of the contributions to three of the CHAEA learning styles. The tendency towards the remaining learning style is implicitly shown in the size and color shades of the points (bigger and darker colors imply a larger tendency). The principal directions shown as continuous lines (PC0 , red; PC1, green; PC2, purple; PC3, orange) emerge from centroid, which is given by the average mean of the data set and is shown as a black star. The projection of the points and the average mean is shown in gray. The projection confidence interval of the average mean is also shown. The origin, which is defined as (0, 0, 0, 0) in the 4D space of the learning styles is shown as a black triangle. The colored points correspond to the maximal pure learning styles (e.g., (20, 0, 0, 0) for activist (red), or (0, 20, 0, 0) for reflector (green) are also marked.

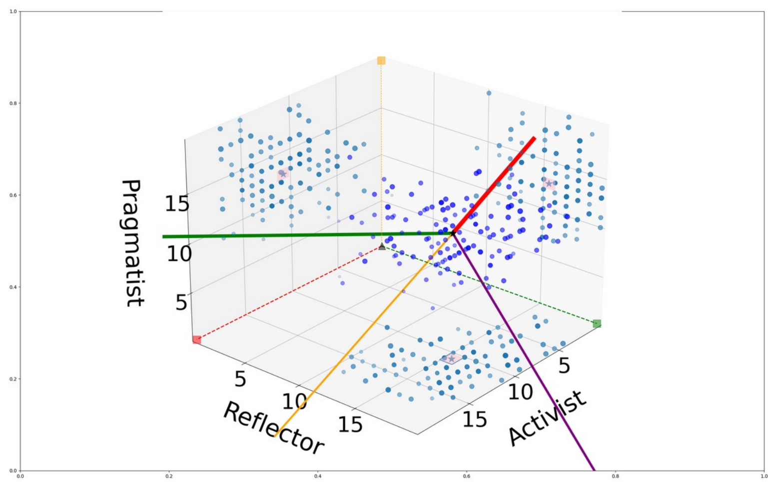

Figure 8. Four-dimensional representation of the learning styles of the students as a function of the CHAEA learning styles.

The central panel in Fig. 9 shows the projection of the learning styles of the student represented in Fig. 8 on the two main principal components (PC0 and PC1), along with their histograms. The data surround the average mean (gray star). The points (20, 0, 0, 0), (0, 20, 0, 0), (0, 0, 20, 0), and (0, 0, 0, 20), which correspond, respectively, to the maximal pure activist (red), reflector (green), theorist (purple), and pragmatist (orange) learning styles in the original basis set are also marked, and joined with a dashed line to the corresponding origin (0, 0, 0, 0) (black triangle). The point (20, 20, 20, 20) is also shown (brown diamond).

The reduced dimensional representation of the data given by Fig. 8 is usually meaningful if Σ_1(%)≥70.0%; in the case under study we have that Σ_1(%)=92.0. The top and left panels of Fig. 9 show the histrograms of the data as a function of PC0 (red) and PC1 (green), which are fitted using Weibull distributions with the parameters listed in Table 11.

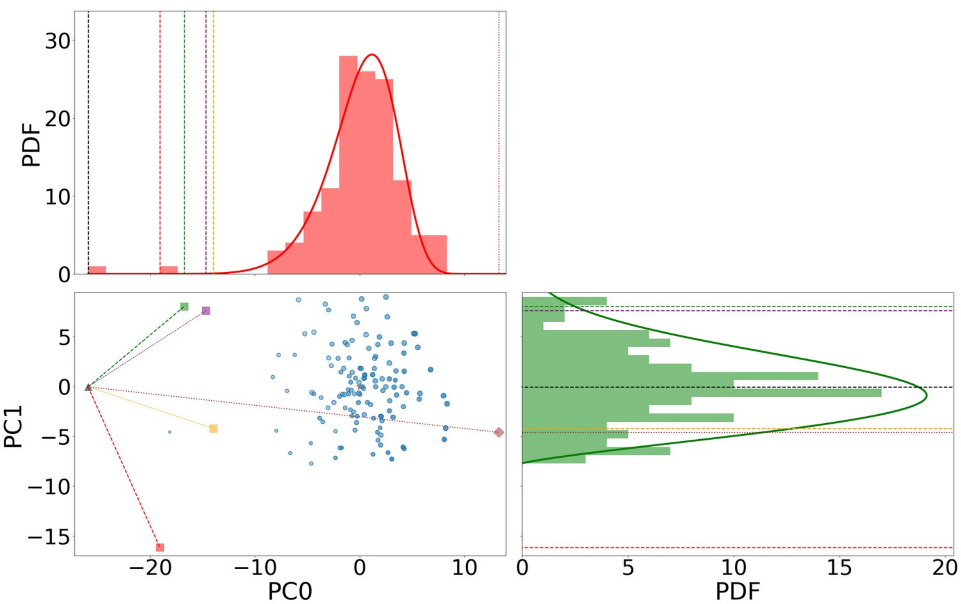

Figure 9. The projection of the learning styles of the students represented in Fig. 8 on the two main principal components (PC0 and PC1), along with their histograms as a function of PC0 (red) and PC1 (green).

Table 11. Parameters of the Weibull distributions that fit the probability distributions (histograms) of the projection of the learning styles on the principal-components basis set (see histograms in Fig. 9). The location parameter θ is set equal to the smallest projection for each component.

Principal component	α	k	θ
0	27.4	9.54	-25.92
1	8.87	2.25	-7.72
2	6.74	3.03	-5.96
3	7.51	4.95	-6.99

2.4 Relation between the learning styles of the students and of the professors in the principal-component space

Figures 10 to 12 show the projection of the learning styles of the students on the two main principal components (PC0 and PC1), as represented in the central panel of Fig. 9, along with the results for the professors (gray crosses). The students with the same qualitative tendency as the professors have been represented with colored crosses.

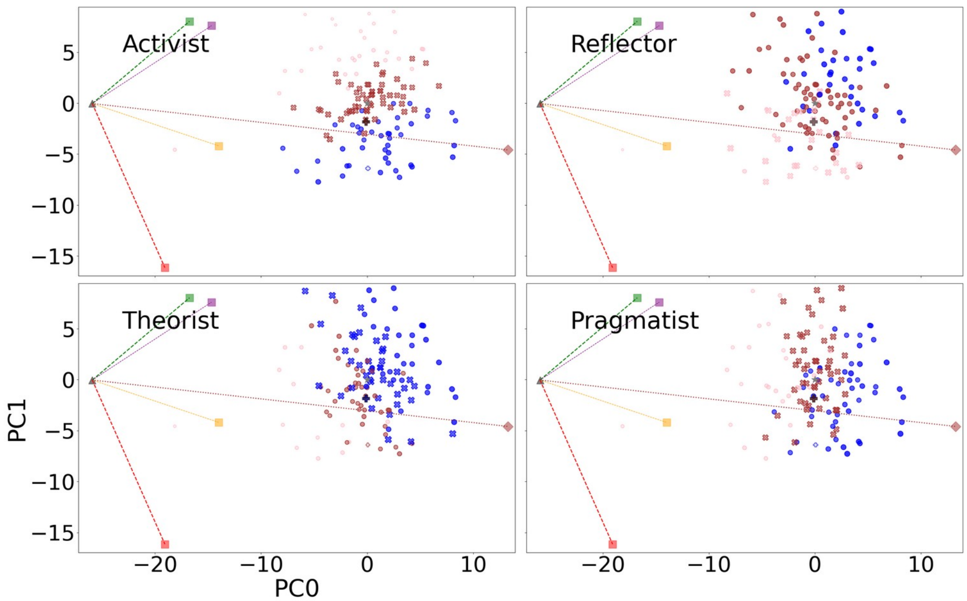

Figure 10. Projection of the learning styles of the students on the two main principal components (PC0 and PC1), as represented in the central panel of Fig. 9 along with the results of the professor of Chemistry (gray cross). The colored crosses mark the students with the same qualitative tendency towards the corresponding learning style (light shade for very low and low tendencies, medium shade for moderate tendency, and dark shade for high and very high tendencies).

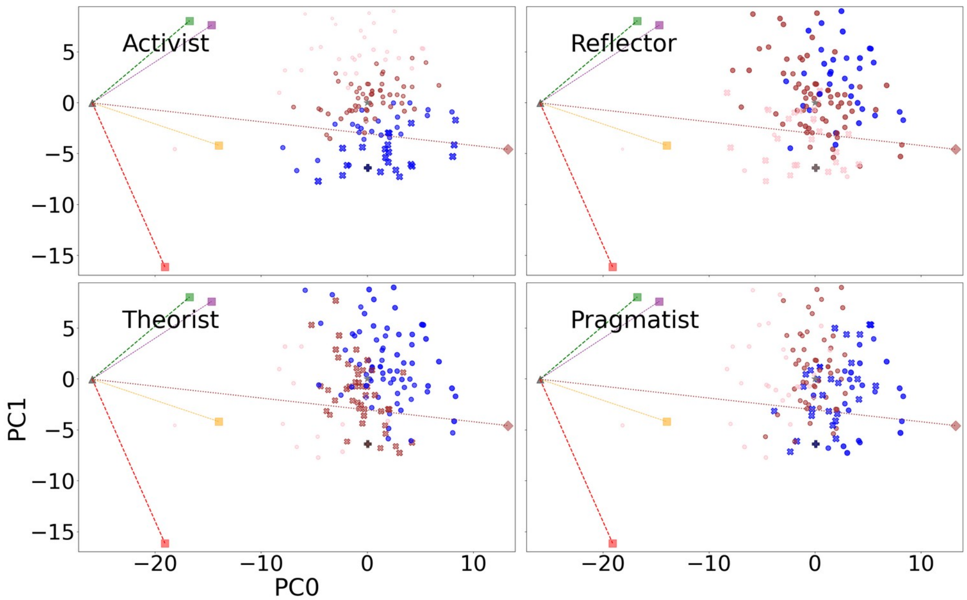

Figure 11. Same as Fig. 10 for the professor of Physics.

3. Participation ratios

Third, the participation ratios are presented in this section. The values for the CHAEA (activist, reflector, theorist, and pragmatist), and for the principal-components basis sets are listed. A statistical analysis of the distribution of the participation ratios is also included.

3.1 Individual analysis of the participation ratios

Table 12 shows the participation ratios of each individual student for CHAEA (activist, reflector, theorist, and pragmatist) and for the principal-components basis sets. For the case under study, this parameter lies between 1 and 4. The smaller the participation ratio, the better.

Table 12. Participation ratios of each individual student for CHAEA learning styles (LS) and for the principal-components (PC) basis sets.

Student	LS	PC
fisicaI_2022-23_01	3.98	2.02
fisicaI_2022-23_02	3.31	1.18
fisicaI_2022-23_03	3.94	2.32
fisicaI_2022-23_04	2.96	3.58
fisicaI_2022-23_05	3.63	2.51
fisicaI_2022-23_06	3.71	2.33
fisicaI_2022-23_07	3.5	1.8
fisicaI_2022-23_08	3.21	2.47
fisicaI_2022-23_09	3.22	1.68
fisicaI_2022-23_11	3.85	1.86
fisicaI_2022-23_12	3.89	1.95
fisicaI_2022-23_13	3.84	1.79
fisicaI_2022-23_14	2.31	2.63
fisicaI_2022-23_15	3.8	1.08
fisicaI_2022-23_17	3.48	2.21
fisicaI_2022-23_18	3.19	2.56
fisicaI_2022-23_19	3.65	2.48
fisicaI_2022-23_20	3.86	2.03

fisicaI_2022-23_21	3.77	1.83
fisicaI_2022-23_22	3.42	2.22
fisicaI_2022-23_23	3.44	3.23
fisicaI_2022-23_24	3.59	1.65
fisicaI_2022-23_25	3.7	3.15
fisicaI_2022-23_26	3.61	2.43
fisicaI_2022-23_27	3.65	1.59
fisicaI_2022-23_30	3.49	1.32
fisicaI_2022-23_31	3.34	3.32
fisicaI_2022-23_33	3.56	2.77
fisicaI_2022-23_34	3.21	2.48
fisicaI_2022-23_35	2.9	1.92
fisicaI_2022-23_36	3.63	1.6
fisicaI_2022-23_37	3.48	3.34
fisicaI_2022-23_38	2.86	1.56
fisicaI_2022-23_39	2.81	1.17
fisicaI_2022-23_40	3.03	1.47
fisicaI_2022-23_41	3.75	1.92
fisicaI_2022-23_42	3.58	3.93
fisicaI_2022-23_43	3.54	1.37
fisicaI_2022-23_44	3.89	1.8
fisicaI_2022-23_45	3.89	2.21
fisicaI_2022-23_46	3.04	2.86
fisicaI_2022-23_47	3.5	1.82
fisicaI_2022-23_48	3.24	2.11

fisicaI_2022-23_49	3.75	1.97
fisicaI_2022-23_50	2.77	1.03
fisicaI_2022-23_51	2.4	1.77
fisicaI_2022-23_52	3.08	1.81
fisicaI_2022-23_53	3.67	1.62
fisicaI_2022-23_54	3.58	3.93
fisicaI_2022-23_55	3.58	1.51
fisicaI_2022-23_56	3.16	2.02
fisicaI_2022-23_57	3.95	1.78
fisicaI_2022-23_58	3.64	2.67
fisicaI_2022-23_59	2.78	2.04
fisicaI_2022-23_61	3.92	2.13
fisicaI_2022-23_62	3.42	2.79
fisicaI_2022-23_63	3.52	2.97
fisicaI_2022-23_64	3.06	1.1
fisicaI_2022-23_65	3.54	1.32
fisicaI_2022-23_66	2.51	1.52
fisicaI_2022-23_67	3.78	2.1
fisicaI_2022-23_68	3.54	3.37
fisicaI_2022-23_69	3.85	1.35
fisicaI_2022-23_70	2.94	2.1
fisicaI_2022-23_71	1.0	1.08
fisicaI_2022-23_72	3.71	2.52
fisicaI_2022-23_73	3.76	1.38
fisicaI_2022-23_74	2.73	2.54
fisicaI_2022-23_75	3.79	2.61

fisicaI_2022-23_76	3.26	2.01
fisicaI_2023-24_01	3.63	2.71
fisicaI_2023-24_02	3.51	1.78
fisicaI_2023-24_03	3.85	1.36
fisicaI_2023-24_04	3.71	1.83
fisicaI_2023-24_05	2.94	2.07
fisicaI_2023-24_06	3.23	2.11
fisicaI_2023-24_07	3.3	1.85
fisicaI_2023-24_08	3.52	1.31
fisicaI_2023-24_09	3.91	2.57
fisicaI_2023-24_10	3.41	2.1
fisicaI_2023-24_11	3.18	1.26
fisicaI_2023-24_12	3.54	1.32
fisicaI_2023-24_13	2.86	2.41
fisicaI_2023-24_14	3.79	2.01
fisicaI_2023-24_15	3.04	2.41
fisicaI_2023-24_16	3.14	1.59
fisicaI_2023-24_17	3.48	2.15
fisicaI_2023-24_18	3.47	2.86
fisicaI_2023-24_20	2.8	1.19
fisicaI_2023-24_21	3.09	1.07
fisicaI_2023-24_22	3.22	2.98
fisicaI_2023-24_23	3.52	2.59
fisicaI_2023-24_24	3.51	1.7
fisicaI_2023-24_25	3.77	2.24
fisicaI_2023-24_26	2.98	1.54

fisicaI_2023-24_27	3.92	1.11
fisicaI_2023-24_28	2.92	1.3
fisicaI_2023-24_29	2.51	1.35
fisicaI_2023-24_30	3.84	2.78
fisicaI_2023-24_31	3.83	1.85
fisicaI_2023-24_32	3.25	1.77
fisicaI_2023-24_33	3.8	2.89
fisicaI_2023-24_34	3.76	2.22
fisicaI_2023-24_35	3.18	1.42
fisicaI_2023-24_36	3.83	1.46
fisicaI_2023-24_37	3.41	2.38
fisicaI_2023-24_38	3.23	2.06
fisicaI_2023-24_39	3.71	2.4
fisicaI_2023-24_40	3.42	3.09
fisicaI_2023-24_41	2.76	1.75
fisicaI_2023-24_42	3.7	1.19
fisicaI_2023-24_43	3.07	1.84
fisicaI_2023-24_44	2.59	1.32
fisicaI_2023-24_45	3.37	3.07
fisicaI_2023-24_46	2.58	1.3
fisicaI_2023-24_47	3.73	1.07
fisicaI_2023-24_48	3.69	1.16
fisicaI_2023-24_49	3.59	1.86
fisicaI_2023-24_50	2.99	1.09
fisicaI_2023-24_51	3.45	2.33
fisicaI_2023-24_52	3.81	2.15

fisicaI_2023-24_53	3.6	1.47
fisicaI_2023-24_54	3.82	2.7
fisicaI_2023-24_55	3.71	2.61
fisicaI_2023-24_56	3.7	1.48
fisicaI_2023-24_57	3.73	3.1
fisicaI_2023-24_58	3.4	1.9
fisicaI_2023-24_59	3.88	1.13
fisicaI_2023-24_60	3.86	1.06

3.2 Global analysis of the participation ratios

Figure 12 shows the probability distribution functions of the participation ratios for the learning-styles basis set (red) and for the principal components (blue). The vertical dashed lines mark the average means along with their corresponding confidence intervals (shaded areas), whose values can be found in Table 13. The corresponding cumulative distributions used in the fitting are shown in Fig. 13.

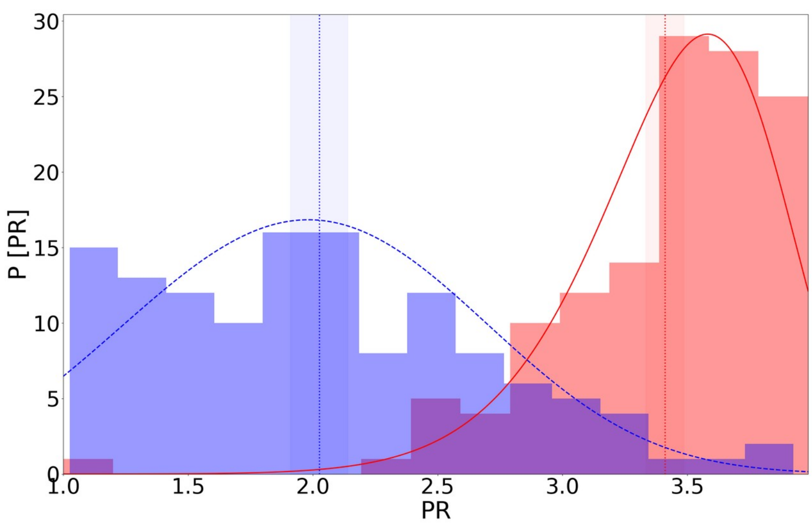

Figure 12. Probability distribution functions of the participation ratios associated with the learning-styles basis set (red) and with the principal components (blue). The vertical dashed lines mark the average means along with their corresponding confidence intervals (shaded areas), whose values can be found in Table 13. The continuous lines show fitting functions given by Weibull distributions with parameters shown in Table 13.

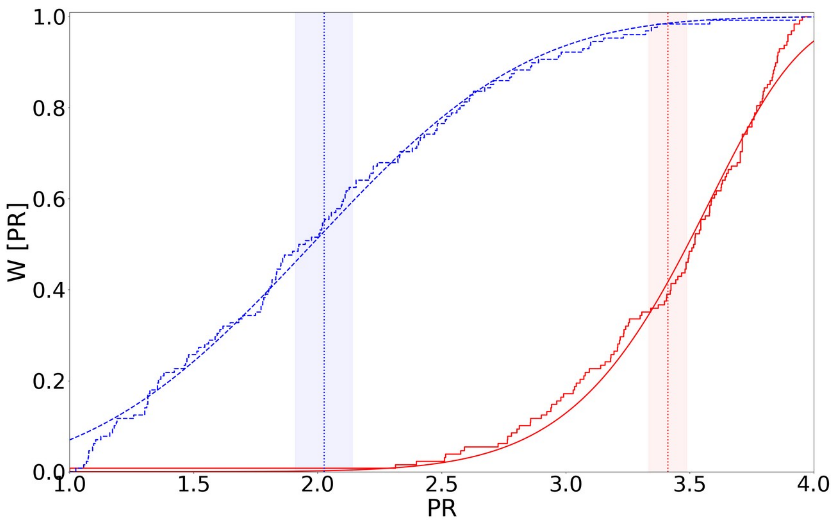

Figure 13. Cumulative distributions for the results of Fig. 12.

Table 13. Average mean and corresponding uncertainty (in parenthesis) of the participation ratios for the learning-styles basis set and for the principal components. α and k are, respectively, the shape and scale parameters of the Weibull distributions the fit the probability distributions (histograms) of Fig. 13. The location parameter is set equal to θ = 0.

Basis set	Average mean (Uncertainty)	α	k
Learning styles	3.41(0.07)	3.62	10.62
Principal components	2.03(0.11)	2.21	3.31

4. Analysis of the students marks

This section is devoted to the analysis of the students marks contained in the input folder marks_K2.

The tables present the averages and uncertainties (in parenthesis) of the learning styles And the principal components (x) and the marks (y) of the whole group, along with the parameters of the linear regressions that fit the marks as a function of the number of points in the corresponding learning style or principal components, being m the slope, b the intercept, and r the regression parameter. Recall that the closest the regression parameter to -1 or 1, the better the fitting. Conversely, a regression parameter close to zero indicates the lack of correlations between the marks and the corresponding variable.

The results marked as nan indicate failures in the linear regressions because, e.g., having too few points.

4.1 Marks as a function of CHAEA learning styles

In this section, we report on the relationship between the marks and CHAEA learning styles.

Marks of the document marks_2022-23.xls

Marks of the column CLASSROOM of the sheet Physics

No clustering analysis based on K-means algorithm is performed due to the low number of students for the marks contained in the column CLASSROOM of the sheet Physics of the document marks_2022-23.xls. Consequently, no figures or tables relating the marks and the CHAEA punctuations for the different learning styles are included here.

Marks of the column LABORATORY of the sheet Physics

No clustering analysis based on K-means algorithm is performed due to the low number of students for the marks contained in the column LABORATORY of the sheet Physics of the document marks_2022-23.xls. Consequently, no figures or tables relating the marks and the CHAEA punctuations for the different learning styles are included here.

Marks of the column EXAM 1 of the sheet Physics

No clustering analysis based on K-means algorithm is performed due to the low number of students for the marks contained in the column EXAM 1 of the sheet Physics of the document marks_2022-23.xls. Consequently, no figures or tables relating the marks and the CHAEA punctuations for the different learning styles are included here.

Marks of the column EXAM 2 of the sheet Physics

No clustering analysis based on K-means algorithm is performed due to the low number of students for the marks contained in the column EXAM 2 of the sheet Physics of the document marks_2022-23.xls. Consequently, no figures or tables relating the marks and the CHAEA punctuations for the different learning styles are included here.

Marks of the column FINAL of the sheet Physics

No clustering analysis based on K-means algorithm is performed due to the low number of students for the marks contained in the column FINAL of the sheet Physics of the document marks_2022-23.xls. Consequently, no figures or tables relating the marks and the CHAEA punctuations for the different learning styles are included here.

Marks of the column CLASSROOM of the sheet Physics

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Marks of the column EXAM 2 of the sheet Chemistry

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Marks of the document marks_2023-24.xls

Marks of the column CLASSROOM of the sheet Physics

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4.2 Marks as a function of the principal components

In this section, we report on the relationship between the marks and the principal components.

Marks of the document marks_2022-23.xls

Marks of the column CLASSROOM of the sheet Physics

No clustering analysis based on K-means algorithm is performed due to the low number of students for the marks contained in the column CLASSROOM of the sheet Physics of the document marks_2022-23.xls. Consequently, no figures or tables relating the marks and the CHAEA punctuations for the different learning styles are included here.

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5. Students characteristics and recommendations

This section summarizes some of the characteristics associated with the activist, theorist, reflector, and pragmatist learning styles, and provides some recommendation to increase the tendencies towards them, as detailed by Alonso, Gallego & Honey (2007), and Cancino (2014).

Alonso, C. M., Gallego, D. J., & Honey, P. (2007). Los estilos de aprendizaje. Procedimiento de diagnóstico y mejora (7th ed.). Ediciones Mensajero, S. A. U.

Cancino, O. (2024, March). Blog de la prof de Français: espacio para intercambiar con los estudiantes. add_hyperlink(p, "https://oticar.wordpress.com/estilos-de-aprendizaje/", "https://oticar.wordpress.com/estilos-de-aprendizaje/"

5.1 Affinity and dominant learning style

According to the affinity, the dominant learning style is the theorist. The value of its affinity is equal to 89.1 %.

The main characteristics of the theorist learning style are:

- Methodical
- Logical
- Objective
- Critical
- Structured

Furthermore, other characteristics usually associated with the theorist learning style are being: disciplined, planned, systematic, organized, synthetic, reasoner, thinker, connector, perfectionist, generalizer, hypothesis-seeker, theory-seeker, model-seeker, question-asker, assumption-explorer, concept-seeker, clear-purpose seeker, rationality-seeker, "why"-asker, value-system and criteria-seeker, procedure-inventor, explorer.

According to the principal components analysis, the differences of the students' learning styles mostly correspond to the pragmatist learning style, as 35.6 % of the eigenvector associated with the main principal component (PC0) is projected onto it.

The main characteristics of the pragmatist learning style are:

- Experimenter
- Practical
- Direct
- Efficient
- Realistic

Furthermore, other characteristics usually associated with the pragmatist learning style are being: technical, useful, fast, decisive, planner, positive, concrete, objective, clear, self-confident, organizer, contemporary, problem-solver, applier of learning, action planner.

Activist learning style

The activist learning style has an average and an uncertainty (in parenthesis) of 10.8(0.6) points.

People with scores higher than 13 on the test find more comfortable in learning situations where they can:

- Attempt new things, new experiences, new opportunities.
- Compete in a team.
- Generate ideas without formal or structural limitations.
- Solve problems.
- Change and vary things.
- Address multiple tasks.
- Engage in dramatization.
- Play roles.
- Experience situations of interest or crisis.
- Capture attention.

Similarly, they'll feel uncomfortable in situations that demand:

- Presenting topics with a heavy theoretical load: Explaining causes, backgrounds...
- Assimilating, analyzing, and interpreting many unclear data.
- Paying attention to details.
- Working alone, reading, writing, or thinking alone.
- Evaluating in advance what is going to be learned.
- Assessing what has already been achieved or learned.
- Repeating the same activity.

If someone has scored below 9, it's advisable to develop certain aspects that characterize this learning style further. Suggestions for improvement include:

- Trying something new, something never done before, at least once a week... Bringing something attention-grabbing to the study or workplace. Reading a newspaper with opposing opinions. Rearranging furniture at home or work.
- Practicing initiating conversations with strangers. In large gatherings, parties, conferences, forcing oneself to initiate and sustain conversations with everyone present, if possible. During leisure time, trying to engage in dialogue with strangers or convince them of our ideas.
- Deliberately fragmenting the day by cutting and changing activities every half an hour...

Blocks that inhibit the development of the activist style:

- Fear of failure or making mistakes.
- Fear of ridicule.

- Anxiety towards new or unfamiliar things.
- Strong desire to thoroughly think things through in advance.
- Self-doubt, lack of self-confidence.
- Taking life very seriously, very conscientiously.

Reflector learning style

The reflector learning style has an average and an uncertainty (in parenthesis) of 15.5(0.5) points.

People with scores higher than 18 on the test learn better in learning situations where they can:

- Observe.
- Reflect on activities.
- Exchange opinions with others by prior agreement.
- Reach decisions at their own pace.
- Work without pressures or mandatory deadlines.
- Review what's been learned, what has happened.
- Investigate thoroughly.
- Gather information.
- Probe to get to the heart of the matter.
- Think before acting.

Similarly, they'll feel uncomfortable in situations that demand:

- Taking the forefront.
- Acting as a leader.
- Chairing meetings or debates.
- Performing in front of observing individuals.
- Playing a role.
- Participating in situations that require action without planning.
- Doing something without prior notice. Presenting an idea spontaneously.

If someone has scored below 14, they might consider the following suggestions to enhance the reflector style:

- Practice observation. Study people's behavior. Note who speaks the most, who interrupts, how often the teacher summarizes... Study non-verbal behavior, when people look at the clock, cross their arms, bite their pencil...
- Keep a personal diary. Reflect on the day's events and see if any conclusions can be drawn from them.
- Practice review after a meeting or event. Review the sequence of events, what went well, what could be improved. Record a dialogue or conversation on a tape recorder and play it back at least twice. Make a list of lessons learned in this way.

Blocks that hinder the development of the reflector style:

- Not having enough time to plan and think.

- Preferring or enjoying quickly changing from one activity to another.
- Being impatient to start action.
- Having resistance to listening carefully and analytically.
- Resistance to presenting things in writing.

Theorist learning style

The theorist learning style has an average and an uncertainty (in parenthesis) of 13.4(0.5) points.

People with scores higher than 14 on the test prefer learning opportunities where they can:

- Feel in structured situations with a clear purpose.
- Fit all data into a system, model, concept, or theory.
- Have time to methodically explore associations and relationships between ideas, events, and situations.
- Have the chance to question.
- Participate in a question-and-answer session.
- Test methods and logic that form the basis of something.

At the same time, they'll avoid learning situations where they have to:

- Be forced to do something without a clear context or purpose.
- Participate in situations where emotions and feelings predominate.
- Engage in unstructured activities with uncertain or ambiguous purposes.
- Take part in open-ended problems.
- Act or decide without a foundation of principles, concepts, policies, or structure.

If someone has scored below 10, they might want to consider the following suggestions to enhance the theorist style:

- Read something dense that provokes thinking for 30 minutes each day (Logic, Linguistics, Theories...). Then try to summarize what they have read using their own words.
- Practice detecting inconsistencies or weaknesses in other people's arguments, in reports, in newspaper articles... Take two newspapers with different ideologies and regularly perform a comparative analysis of the differences in their perspectives.
- Take a complex situation and analyze it to pinpoint why it developed that way, what could have been done differently, and at what point. Historical or everyday life situations. Analyze how they've used their own time...

Blocks that hinder the development of the theorist style:

- Being swayed by first impressions.
- Preferring intuition and subjectivity.
- Disliking structured and organized approaches.
- Preferring spontaneity and risk-taking.

Pragmatist learning style

The pragmatist learning style has an average and an uncertainty (in parenthesis) of 12.7(0.5) points.

People with scores higher than 16 on the test find more comfortable in situations that allow them to:

- Learn techniques for doing things with evident practical advantages.
- Be exposed to a model they can emulate.
- Acquire immediately applicable techniques in your work.
- Have the immediate possibility to apply what they've learned, to experiment.
- Develop action plans with a clear result.
- Give directions, suggest shortcuts.

Likewise, they'll find difficulty in situations that demand:

- Realizing that learning isn't related to an immediate need that they recognize or cannot see.
- Perceiving that the learning doesn't have immediate importance or practical benefit.
- Learning something that is distant from reality.
- Learning theories and general principles.
- Working without clear instructions on how to do it.

If someone has scored below 10, the following suggestions will be useful for enhancing the pragmatist style:

- Gather techniques, practical ways of doing things. They can cover anything that might be useful: analytical techniques, interpersonal skills, assertiveness, time-saving presentation techniques, statistics, memory improvement techniques...
- Seek help from individuals with demonstrated experience. They can observe our technique and advise on how to improve it.
- Focus on developing action plans in meetings and discussions of all kinds. These action plans should be specific and have deadlines. Make it a rule never to leave a meeting, debate, or class without a list of actions for yourself.

Blocks that hinder the development of the pragmatist style:

- Interest in the perfect solution rather than practicality.
- Considering useful techniques as exaggerated simplifications.
- Always leaving topics open and not committing to specific actions.
- Believing that others' ideas won't work when applied to their situation.
- Enjoying marginal topics or getting lost in them.

If you have any comments or suggestions, feel free to reach out by sending an email with your feedback to fabio.revuelta@upm.es.